RMO(ORISSA) – 2006

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer as many questions as you can.

1.
   (a) One packet contain 5 numbers of thing A, 3 numbers of thing B and 7 numbers of thing C and its costs is Rs. 24.50 but if packet contain 2 numbers of thing A, 2 number of thing B and 3 numbers of thing C and its costs is Rs. 17. Find the cost of packet if it contain 16 numbers of thing A, 9 numbers of thing B and 23 numbers of thing C.

   (b) Tow tapes A and B fill one tank in $7\frac{1}{2}$ and 5 minutes respectively. Third tape C, empty 14 litres of water in every one minute. If in that full tank, we open all the three tapes at a time, tank will be empty in 1 hour. Find the capacity of tank.

   (c) If any number when divided by 48 after subtracting 3 from the number gives quotient an odd number and remainder 5. Find the remainder when that number is divided by 32.

   (d) Let abc is one 3 digit number such that $abc = (bc)^2$ and $bc = c^2$ where bc is 2 digit number. Find a, b, c.

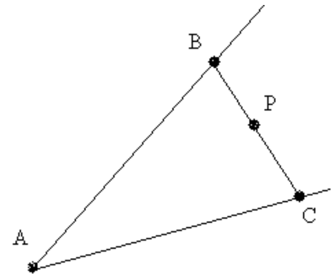
   (e) Find the unit digit after simplification of $4444^{4001} - 1993^{1890}$.

2.
   (f) If m and n are positive integers and m is an odd number then find greatest common divisor of $2^m - 1$ and $2^n + 1$.

   (g) To count numbers of fishes in a pond One scientist follows like this: First day he caught 30 fishes color them and kept in the same pond. Second day again he caught 40 fishes which contains 2 colored fish. If colored fishes were perfectly distributed in the pond. Then find the number of fishes in the pond.

3. Prove that $3^{105} + 4^{105}$ is divisible by 13, 49 and 181 but not divisible by 5 or 11.
4. Represent $2^{2040} - 1$ as a product of six factors such that each factor is greater that 2^{254}.
5. Solve $x = \sqrt{a - \sqrt{a + x}}$, when $a \geq 1$ and take positive value of $\sqrt{\ }$ only.
6. If $a_1, a_2, ..., a_n$ are positive real numbers and $S = a_1 + a_2 + ... + a_n$, then prove that $\frac{S}{S-a_1} + \frac{S}{S-a_2} + ... + \frac{S}{S-a_n} \geq \frac{n^2}{n-1}$.
7. If x_0 is a root of polynomial $p(x_0) = 0$ i.e. $p(x) = a_0 + a_1x + a_2x^2 + ... + a_nx^n$, then find all roots of $p_n(x) = 1 + x + \frac{x(x+1)}{2!} + \frac{x(x+1)(x+2)}{3!} + ... + \frac{x(x+1)(x+2)....(x+n-1)}{n!}$.
8. In triangle ABD, AB = BC, Altitude AD intersect circum circle at P. Prove that $AP.BC = 2AB.BP$.
9. In triangle ABC, D and E are two points on Ac and AB respectively. BD and CE intersect at X, such that $\frac{AD}{DC} = \frac{1}{2}$ and $\frac{BE}{AE} = \frac{1}{3}$. Prove that area of triangle AXC is two-third of area of triangle ABC.

10. In the given figure how to draw BPC through P such that value of $\frac{1}{BP}+\frac{1}{PC}$ is maximum.



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